

The Copenhagen–Everett Bridge: Information Conservation Principle Unifies Quantum Interpretations

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Abstract

The Copenhagen and Everett interpretations of quantum mechanics have been treated as competitors for nearly seven decades. Copenhagen posits wavefunction collapse producing one definite outcome. Everett posits branching producing all outcomes. The Born rule — the probability measure $P(a) = |\langle a | \psi \rangle|^2$ — is the wedge that keeps them apart: Copenhagen treats it as an unexplained postulate; Everett derives it from rationality axioms that remain contested.

This paper proposes that the wedge can be removed by a universal conservation law. The Information Conservation Principle (ICP) — which formalizes information conservation as a first-class physical principle — constrains both interpretations to assign the same unique probability measure. The conserved quantity is identified as the squared-amplitude measure of the universal state, interpreted as information capacity (distinguishable state-space volume). By Busch's extension of Gleason's theorem, the Born rule is the unique probability measure on quantum state space consistent with this conserved measure.

Three theorems close the derivation: (1) the C^* -algebra structure is **derived** from operational postulates via the ICP rather than assumed; (2) the 2-norm is shown to be the unique unitarily-invariant, no-signaling-compatible conserved scalar on Hilbert space; and (3) the capacity interpretation is grounded in a uniqueness result showing that the 2-norm is the only permutation-invariant, subdivision-additive, normalized functional of the branch distribution.

The result is a bridge rather than a competitor: Copenhagen preserves one outcome per branch locally, requiring no modification. Everett preserves the branch ensemble globally, requiring no modification. The ICP constrains them both, ensuring that the Born rule probabilities are physically necessary rather than interpretation-dependent. Probability-Weighted Branch Realism (PWBR) provides the physical interpretation: branches are real but probability-weighted, with the Born rule measure grounded in information conservation rather than postulated.

The measurement problem in quantum mechanics has two layers. The collapse question — what converts a superposition into a definite outcome — separates Copenhagen from Everett. The probability question — why the outcomes follow $P(a) = |\langle a|\psi \rangle|^2$ — has been treated as a separate puzzle, with each interpretation offering a different account.

Copenhagen treats the probabilities as postulates, adding the Born rule to the unitarity axioms without physical explanation. Everett attempts to derive them from branch counting via decision-theoretic arguments (Deutsch 1999; Wallace 2012), but these derivations have been contested on grounds that the rationality axioms either assume what they derive or are not uniquely motivated (Lewis 2010; Baker 2007). Zurek's envariance derivation (2005) has been criticized for presupposing the tensor product structure that itself implies Born-rule statistics (Schlosshauer & Fine 2005).

This paper takes a different approach. Rather than deriving the Born rule from rationality or symmetry within a given interpretation, it derives the Born rule — and the Hilbert space structure it lives on — from operational information-theoretic axioms that are interpretation-neutral. The reconstruction follows Chiribella, D'Ariano & Perinotti (2011) [CDP], with the capacity/content distinction clarifying the role of the 2-norm. The wedge between Copenhagen and Everett is removed not by choosing one interpretation, but by showing that both are special cases of a formalism in which the Born rule is a theorem.

Hardy, L. (2001). Quantum theory from five reasonable axioms. arXiv:quant-ph/0101012.

Masanes, L., & Müller, M. P. (2011). A derivation of quantum theory from physical requirements. *New Journal of Physics*, 13(6), 063001.

Section 2 states the operational primitives and axioms (replacing the previous ICP postulate). Section 3 formalizes the 2-norm as information capacity within the derived Hilbert space. Section 3a presents three theorems that close the derivation from first principles. Section 4 presents the bridge itself. Section 5 gives the mathematical uniqueness argument. Section 6 develops PWBR. Section 7 discusses implications for the DPH framework. Section 8 states honest limitations. Section 9 concludes.

Section 2 states the Information Conservation Principle. Section 3 formalizes the 2-norm as information capacity. Section 3a presents three theorems that close the derivation. Section 4 presents the bridge itself. Section 5 gives the mathematical uniqueness argument. Section 6 develops PWBR. Section 7 discusses implications for the DPH framework. Section 8 states honest limitations. Section 9 concludes.

2. Operational Primitives and Axioms

The previous version of this framework (Varney, 2025a) postulated an Information Conservation Principle (ICP) identifying the 2-norm $\sum |a_i|^2$ as a conserved informational quantity. As critics correctly noted, this is circular: the 2-norm *is* the Born rule weights, so postulating its

conservation assumes the conclusion. This section replaces that postulate with a non-circular foundation.

We begin with minimal operational primitives, following Chiribella, D'Ariano & Perinotti (2011) [CDP] and the general framework of Barrett (2007):

- Systems: physical entities that can be prepared, transformed, and measured.
- Tests: collections of outcomes $\{i\}$ with associated probabilities $p(i|\omega, T)$ for state ω under test T .
- Composition: systems A and B compose into AB ; tests compose in sequence and in parallel. No Hilbert space, no C^* -algebra, no inner product is assumed. This is a convex operational theory.

The axioms (CDP, 2011) are:

Axiom 1 (Purification). Every state of a system arises as the marginal of a pure state of a larger system, and this purification is essentially unique up to reversible transformations on the purifying system.

Axiom 2 (Ideal Compression). For any state, there exists an encoding into a smaller system that perfectly preserves the state's statistics, and this encoding is reversible.

Axiom 3 (Tomographic Locality). The state of a composite system is completely determined by the statistics of local measurements on the components.

Axiom 4 (Continuous Reversibility). Between any two pure states there exists a continuous reversible transformation.

These axioms are operationally testable and make no reference to Hilbert space or the Born rule.

From these axioms, the CDP reconstruction proves: the state space of any system is the cone of positive semidefinite operators on a complex Hilbert space; tests are POVMs; reversible transformations are unitary conjugations; composition is the tensor product. The 2-norm $\sum |a_i|^2 = \langle \Psi | \Psi \rangle$ emerges as the unique conserved scalar under the derived unitary evolution — not as a postulate, but as a theorem.

The capacity/content distinction remains central. In the derived formalism:

- Capacity: the 2-norm $C(\Psi) = \langle \Psi | \Psi \rangle = 1$, conserved because unitary evolution preserves the inner product.
- Content: von Neumann entropy $S(\rho) = -\text{Tr}(\rho \ln \rho)$, which measures mixedness/uncertainty and is not conserved under measurement.

The Born rule connects them: probability = capacity allocated to outcome branches.

Definition (Informational measure, derived). In the reconstructed framework, information capacity is the measure of distinguishable state-space volume. For a system with Hilbert space H , the maximum capacity is $\dim(H)$. The total capacity of a state $|\Psi\rangle = \sum_i a_i |i\rangle$ is $C(\Psi) = \sum_i |a_i|^2 = 1$, quantifying how the accessible state-space volume is distributed across orthogonal subspaces. Unitary evolution conserves $C(\Psi)$ because the total accessible volume is invariant — branching partitions it but does not create or destroy it.

3. The 2-Norm as Information Capacity

With the Hilbert space structure derived in Section 2, the 2-norm emerges as the unique conserved scalar. This section develops its interpretation as information capacity, connecting it to established information-theoretic quantities.

3.1 Related Work

The use of the squared-amplitude measure as a branch weight has extensive precedent. Everett (1957) defined the branch measure as the squared norm of the relative state. Brukner & Zeilinger (1999) identified the 2-norm $\sum |a_i|^2$ as an operationally invariant informational quantity. Deutsch (1999) and Wallace (2012) used squared-amplitude weights in decision-theoretic derivations. Carroll & Sebens (2014) developed self-locating uncertainty with squared-amplitude measure.

The present framework differs in three respects. First, the 2-norm is not postulated as conserved — it is derived as the unique unitarily-invariant scalar on the Hilbert space that emerges from operational axioms. Second, the capacity/content distinction clarifies that the 2-norm is conserved because it quantifies state-space volume (capacity), not observer knowledge (content). Third, the non-negativity and normalization conditions for Gleason's theorem are satisfied by the derived 2-norm, not externally imposed.

3.2 Definition

Following Everett (1957), let the universal quantum state be expressed as $|\psi\rangle = \sum_i a_i |i\rangle$. Define the total informational capacity: $C_{total}(\psi) = \sum_i |a_i|^2$, identically 1 for any normalized state. The capacity allocated to branch B_i is $C_i = |a_i|^2$, representing the fraction of the total informational capacity carried by that branch.

3.3 Conservation

Under any unitary transformation U , $|\psi'\rangle = U|\psi\rangle$, the total capacity is invariant: $C_{total}(\psi') = \sum_i |\langle i | U | \psi \rangle|^2 = \langle \psi | U^\dagger U | \psi \rangle = \langle \psi | \psi \rangle = C_{total}(\psi)$. No physical process — unitary evolution, decoherence, branching, or measurement — changes C_{total} . The capacity is only redistributed across degrees of freedom.

3.4 Operational Properties

The capacity measure satisfies three properties that jointly justify its interpretation as an informational resource:

- Subspace additivity. For any orthogonal decomposition $H = H_1 \oplus H_2$, the capacity satisfies $C(P_1\Psi) + C(P_2\Psi) = C(\Psi)$. This is the frame-function condition.
- Measurement independence. For any orthonormal basis $\{|e_i\rangle\}$, the capacity partitions as $\sum_i |\langle e_i | \Psi \rangle|^2 = 1$. The total capacity is independent of basis.
- Asymptotic frequency correspondence. For an ensemble of $N \rightarrow \infty$ identically prepared systems, the squared amplitude $|a_i|^2$ equals the limiting frequency of outcomes associated with branch B_i .

3.5 Capacity vs. Content

Traditional information measures quantify content: Shannon entropy measures uncertainty; von Neumann entropy measures mixedness. Both change under measurement because the observer's knowledge changes. A conserved quantity cannot be a measure of content — content is what gets created and destroyed as observers learn and forget.

Capacity, by contrast, quantifies what the physical state can support, regardless of any observer's knowledge. The Born rule connects the two: an observer's probability for an outcome equals the capacity allocated to the branches yielding that outcome. Capacity is the conserved substrate; probability is the local experience of that substrate by an embedded observer.

3.5a Topological Derivation: Capacity = Topological Entanglement Entropy

The capacity/content distinction is not merely interpretive — it is derived from the topological substrate. In the 4D BF theory substrate (Varney 2026d, §5):

Capacity = Topological Entanglement Entropy γ .

For a spatial region A in the topological phase, the entanglement entropy follows the Kitaev-Preskill / Levin-Wen formula:

$$S_A = \alpha |\partial A| - \gamma + O(1/|\partial A|)$$

where $\gamma = \log D$, $D = \sqrt{\sum_a d_a^2}$ is the total quantum dimension, and d_a are the quantum dimensions of the anyons. The topological term $\gamma = \log D$ is:

- Invariant under local unitary transformations (topological invariant)
- Conserved under all unitary evolution (including branching)
- Additive over independent regions: $\gamma(A \cup B) = \gamma(A) + \gamma(B)$ for disjoint A, B
- Non-negative and normalized: $\gamma \geq 0$, with $\gamma = \log D$ for the maximal region

These are precisely the properties of the capacity measure $C(\Psi) = \sum |a_i|^2 = 1$. The total capacity is $C_{\text{total}} = \log D$ (in nats) or $\log_2 D$ (in bits). The branch capacities $C_i = |a_i|^2$ correspond to the quantum dimension weights d_a^2/D^2 — the Born rule is the quantum dimension distribution.

Content = von Neumann Entropy $S(\rho)$.

Content measures the mixedness/uncertainty of the state ρ . Under measurement, ρ changes $\rightarrow S(\rho)$ changes. Content is created (measurement yields outcome) and destroyed (erasure). It is the Holevo quantity $\chi(E) = S(\rho) - \sum p_i S(\rho_i)$ that bounds extractable classical information.

Why Capacity is Conserved, Content is Not.

- Capacity $\gamma = \log D$ is a topological invariant of the anyon model. It depends only on the anyon types (the modular tensor category), not on the state. Any unitary evolution — including branching — merely redistributes the total quantum dimension among branches: $\sum d_a^2 = D^2$ is invariant.
- Content $S(\rho)$ depends on the state ρ . Measurement changes $\rho \rightarrow \rho_i$, changing $S(\rho)$. The Born rule $p_i = d_a^2/D^2$ connects the two: probability = capacity fraction.

Born Rule from Quantum Dimensions.

The branch weight for anyon type a is $p_a = d_a^2/D^2$. This satisfies:

- Non-negativity: $d_a^2 \geq 0$
- Normalization: $\sum d_a^2 = D^2$
- Additivity under fusion: $N_{\{ab\}}^c d_c = d_a d_b$ (Verlinde formula) $\rightarrow p_a p_b = \sum_c N_{\{ab\}}^c p_c$
- Busch's theorem: any frame function on the anyon Hilbert space has this form.

Thus the Born rule is the quantum dimension distribution — derived from the topological capacity.

3.6 Relationship to the Holevo Bound

The ensemble $E = \{|a_i|^2, |i\rangle\langle i|\}$ has Holevo quantity $\chi(E) = H(\{|a_i|^2\}) = -\sum_i |a_i|^2 \log |a_i|^2$, which bounds the classical information extractable from the branch structure. χ is not conserved: as entanglement redistributes the amplitudes, the Holevo quantity can change. But the measure that generates it — $\sum_i |a_i|^2$ — is conserved. The Shannon/von Neumann entropies are functionals of the capacity distribution, not the capacity itself.

3.7 The 2-Norm as a Resource

The ICP suggests a distinctive resource-theoretic structure: the 2-norm is not merely monotonic but strictly conserved, and the free operations are those that respect this conservation.

Theorem (Unique conserved monotone). The function $C(\Psi) = \sum_i |\langle i|\Psi\rangle|^2 = 1$ is the unique scalar function on H that satisfies: (a) conservation under all free operations; (b) additivity over orthogonal subspaces; (c) normalization: $C(\Psi) = 1$ for all physically allowed states. Uniqueness follows from the fact that the inner product is the unique positive-definite sesquilinear form on H , and C is the induced norm squared.

The resource is never consumed, only redistributed. What does change is the distribution of the 2-norm across orthogonal subspaces (branches), quantified by the Holevo capacity of the branch ensemble. This entropy is not conserved but the total capacity is.

3.7a The Two-Level Architecture of Information Conservation

A critical refinement of the ICP concerns the distinction between conservation laws as experienced by embedded observers and conservation laws as implemented by the computational substrate. Within the DPH framework, the ICP operates at two distinct levels:

Level 1 — Embedded Observer (Absolute ICP). For any observer embedded within a branch of the multiverse, information conservation is absolute. The 2-norm of the local quantum state is strictly conserved; no operation exists to transfer matter, energy, or information between decoherent branches. Time travel, cross-branch communication, and access to the past are fundamentally prohibited — not by technological limitation, but because the requisite operations are undefined within the local algebraic structure. This is the absolute ICP as experienced by all

embedded observers, and it is the form of conservation that underwrites the Born rule, the no-signaling principle, and the thermodynamic arrow of time.

Level 2 — Computational Substrate (Meta-ICP). The computational substrate that instantiates the branch ensemble — the topological quantum error-correcting code described in Section 3.2c and 4.1a — operates under a different constraint. The substrate's objective is not absolute conservation per branch, but the maximization of Novelty Flux Density (NFD) across the ensemble. To achieve the computational efficiency required for a novelty-harvesting multiverse, the substrate employs a cache-coherency protocol: static assets (physical constants, vacuum state, geometric background) are deduplicated across branches, and branch states are cached and shared where NFD is unaffected. This deduplication constitutes a controlled, substrate-level violation of the absolute ICP — but only where it increases Novelty Flux Density without corrupting the harvest.

The leakage operator L derived in Section 4.1a is precisely the cache-coherency protocol of the substrate. When static assets are deduplicated across branches, the cache-coherency protocol propagates updates through the leakage operator L , generating the boundary permeability phenomena described in Varney by the leakage operator. These artifacts — Type I (routing), Type II (observer-state), and Type III (rendering) — are not fundamental violations of the ICP; they are the operational signature of the substrate's cache-coherency protocol. For embedded observers, the ICP remains absolute; the leakage operator represents the substrate's operational boundary, not a violation of the observer's conservation law.

This two-level architecture resolves an apparent tension: the ICP is absolute for embedded observers (supporting the Born rule, no-signaling, and the arrow of time), while the substrate implements a cache-coherency protocol that appears as controlled permeability at the operational boundary. The absolute ICP holds for all embedded physics; the substrate's Meta-ICP — maximize NFD — permits deduplication where it does not corrupt the harvest.

3.8 Hilbert Space from Operational Axioms (Summary of CDP Reconstruction)

The CDP reconstruction (Chiribella, D'Ariano & Perinotti, 2011) derives Hilbert space from the operational primitives and axioms of Section 2. The logical chain is:

Operational primitives (systems, tests, composition) + Purification + Ideal Compression + Tomographic Locality + Continuous Reversibility $\rightarrow$ Complex Hilbert space H , state space = positive semidefinite operators, tests = POVMs, reversible transformations = unitary conjugations, composition = tensor product.

Once the Hilbert space and normalized state exist, the Born rule follows by Busch's extension of Gleason's theorem. The GNS construction is not needed separately — the Hilbert space is directly constructed.

Section 3a presents three theorems that close specific gaps in this derivation.

3a. Three Theorems Closing the Derivation

The CDP reconstruction establishes the Hilbert space structure. The following three theorems close remaining gaps: (1) deriving the C*-algebra structure directly from operational postulates without assuming it; (2) proving the 2-norm is the unique conserved scalar compatible with no-signaling; (3) grounding the capacity interpretation in a uniqueness result for the branch distribution.

Theorem 1: Deriving C*-Algebras from Operational Axioms (CDP Reconstruction)

The Problem

Standard presentations of quantum theory assume C*-algebra structure (observables as self-adjoint operators, states as positive linear functionals). The CDP reconstruction eliminates this assumption by deriving the C*-algebra from operational primitives.

The Fix: Operational Derivation (CDP, 2011; see also Hardy 2001; Masanes & Müller 2011)

We begin with only the following minimal operational primitives:

- Systems: physical entities that can be prepared, transformed, and measured.
- Tests: collections of outcomes $\{i\}$ with associated probabilities $p(i|\omega, T)$.
- Composition: systems A and B compose into AB; tests compose in sequence and parallel.

No Hilbert space, no C*-algebra, no inner product is assumed.

- The CDP axioms (Purification, Ideal Compression, Tomographic Locality, Continuous Reversibility) are stated operationally — they refer only to preparations, transformations, and measurements.
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- Proof Sketch (CDP, 2011, Theorems 1-4):

Step 1: Purification + Ideal Compression force the state space to be self-dual. The cone of states is isomorphic to its dual cone of effects. This implies states can be represented as positive operators.

Step 2: Tomographic Locality forces the composite state space to be the tensor product of component state spaces. Combined with self-duality, this yields the standard quantum composition rule.

Step 3: Continuous Reversibility forces the field to be complex. Real and quaternionic Hilbert spaces fail universal purification or associative composition (Solèr, 1995). Complex Hilbert space is the unique inner-product space over a normed division algebra satisfying all axioms.

Step 4: The C*-algebra $B(H)$ of bounded operators on the derived complex Hilbert space H follows by construction. Observables are self-adjoint elements; states are density operators.

The auxiliary conditions (Purification, Ideal Compression) have direct experimental interpretations and are independently motivated by the no-hiding theorem and Schumacher compression. Neither assumes complex amplitudes or interference.

Theorem 1 (ICP-OP $\rightarrow$ Quantum Theory)

Any convex operational theory satisfying ICP-OP, together with:

1. Purification: every mixed state of system S is the marginal of a pure state of a larger system SE ;
2. Ideal compression: any state can be reversibly compressed to a system whose dimension equals the rank of the state;

is operationally equivalent to finite-dimensional quantum theory. That is, states are density matrices on a complex Hilbert space, reversible transformations are unitary channels, and measurements are POVMs.

The logical chain is: Operational primitives + CDP axioms $\rightarrow$ Complex Hilbert space $H \rightarrow C^*$ -algebra $B(H) \rightarrow$ Normalized states with $C(\Psi) = \|\Psi\|^2 = 1 \rightarrow$ Busch's theorem $\rightarrow$ Born rule uniquely.

Proof Sketch

Step 1: ICP-OP forces the state space to be self-dual. Condition (b) requires the capacity function C to be invariant under the full group G of reversible transformations. Condition (a) requires C to be additive over tensor products. The only scalar function on a convex state space that is both G -invariant and additive over composition is, by a theorem of Alfsen & Shultz (2003), a trace-form: $C(\omega) = \text{Tr}(Q\omega)$ for some positive operator Q . Normalization (c) fixes $Q = I$, giving $C(\omega) = \text{Tr}(\omega)$. This forces the state space to be the cone of positive trace-class operators — i.e., density matrices.

Step 2: Purification + ideal compression force complex Hilbert space. Given that states are density matrices on some inner-product space, purification requires every mixed state to extend to a pure state on a doubled system. For real Hilbert spaces, purification fails generically. For quaternionic Hilbert spaces, composition is not associative. Complex Hilbert space is the unique inner-product space over a normed division algebra for which purification holds universally and composition is associative. This follows from Solèr (1995) combined with the composition requirement.

Step 3: The C^* -algebra structure follows. The bounded operators $B(H)$ on the derived complex Hilbert space H form a C^* -algebra by construction. The GNS argument of Section 3.8 now applies, but crucially the C^* -algebra is derived from ICP-OP and the auxiliary conditions, not assumed.

Regarding the auxiliary conditions: Purification is the operational statement that every probabilistic mixture has a physical explanation in terms of correlations with an environment. This is weaker than assuming unitarity. Ideal compression says redundant degrees of freedom can be removed reversibly. Both postulates have direct experimental interpretations and are independently motivated by the no-hiding theorem and Schumacher compression respectively. Neither assumes complex amplitudes or interference.

The logical chain is now: Operational primitives + ICP-OP + Purification + Ideal compression $\rightarrow$ Complex Hilbert space $\rightarrow C^*$ -algebra $B(H) \rightarrow$ GNS cyclic vector with $C(\Psi) = \|\Psi\|^2 = 1 \rightarrow$ Busch's theorem $\rightarrow$ Born rule uniquely.

Theorem 2: The p-Norm No-Go Theorem (Uniqueness of the 2-Norm)

The Problem

Given a Hilbert space, many p-norms $\sum |a_i|^p$ are mathematically definable. Why is the 2-norm the physically relevant conserved scalar? Without ruling out alternatives, the Born rule derivation is not forced.

The Fix

On a complex Hilbert space, any candidate conserved scalar C_p must satisfy: F1 (Frame function: $\sum C_p(e_i) = 1$ for any orthonormal basis), F2 (Unitary invariance: $C_p(U\Psi) = C_p(\Psi)$), F3 (No-signaling compatibility: marginal states obtained via partial trace are independent of remote measurements).

Theorem 2 (Uniqueness of the 2-norm as informational measure)

Let H be a complex Hilbert space of dimension $d \geq 3$. Suppose a physical theory assigns to each normalized state $|\Psi\rangle = \sum_i a_i |i\rangle$ a conserved capacity measure of the form $C_p(\Psi) = (\sum_i |a_i|^p)^{1/p}$ for some $p > 0$. Then C_p satisfies all three of the following conditions simultaneously if and only if $p = 2$:

1. Frame-function property (F1): for every orthonormal basis $\{e_i\}$, $\sum_i C_p(P_{\{e_i\}}\Psi) = C_p(\Psi)$;
2. Basis independence (F2): $C_p(\Psi)$ is invariant under all unitary transformations U ;
3. No-signaling (F3): the marginal capacity assigned to subsystem A of a bipartite state $|\Psi_{AB}\rangle$ is independent of any measurement performed on B .

Proof

F2 requires $p = 2$ directly. The unit sphere in H is homogeneous under $U(d)$, so C_p must be a function of $\|\Psi\|$ alone. For normalized states, $\|\Psi\|_2 = 1$ always, but $\|\Psi\|_p$ for $p \neq 2$ varies with basis components and is not unitarily invariant. Concretely: $|\Psi\rangle = (1/\sqrt{2})|0\rangle + (1/\sqrt{2})|1\rangle$ and $|\Phi\rangle = |0\rangle$ have $\|\Psi\|_4^4 = 1/2 \neq 1 = \|\Phi\|_4^4$, but a Hadamard maps $|0\rangle$ to $|\Psi\rangle$. The only unitarily-invariant p-norm on normalized vectors is $p = 2$.

F3 eliminates remaining candidates. No-signaling requires that the reduced state on A , obtained by partial trace, depends only on the global state. The partial trace uses the Hilbert-Schmidt inner product, which is sesquilinear — uniquely picking $p = 2$ among p-norms. Linearity of the trace requires this.

F1 follows from $p = 2$. The frame-function property $\sum |\langle e_i | \Psi \rangle|^2 = \|\Psi\|^2 = 1$ is the Parseval identity. For $p \neq 2$, $\sum |\langle e_i | \Psi \rangle|^p$ is basis-dependent.

Corollary. The Born rule is the unique probability measure consistent with any conserved scalar on complex Hilbert space that is unitarily invariant and no-signaling-compatible. The 2-norm is not one choice among many — it is forced by the Hilbert space structure derived from operational axioms.

This theorem means the ICP does not merely permit the Born rule — it forces it. Any post-quantum theory that conserves a different scalar on state space either breaks unitary invariance, permits superluminal signaling, or assigns basis-dependent probabilities. The Born rule is not one choice among many conserved measures. It is the only one.

Theorem 3: Operational Derivation of the Capacity Interpretation

The Problem

The 2-norm is derived as the unique conserved scalar. But why interpret it as 'information capacity' rather than just a mathematical norm? The connection to operational information theory (accessible information, distinguishability) must be derived.

The Fix

Theorem 3 (2-norm as accessible information measure)

Let $E = \{|a_i|^2, |i\rangle\langle i|\}$ be the branch ensemble associated with a measurement on state $|\Psi\rangle = \sum_i a_i |i\rangle$. Then:

1. The Holevo quantity $\chi(E) = H(\{|a_i|^2\})$ is upper-bounded by $\log d$, where d is the dimension of the subspace spanned by the branches.
2. $\chi(E)$ is maximized over all states $|\Psi\rangle$ with n nonzero branches if and only if $|a_i|^2 = 1/n$ for all i .
3. The conserved quantity $C_{\text{total}} = \sum_i |a_i|^2 = 1$ is the unique scalar functional of the branch distribution $\{|a_i|^2\}$ that is: (i) invariant under permutation of branch labels, (ii) additive when branches are subdivided into orthogonal sub-branches, and (iii) equal to 1 for any normalized state regardless of the number of branches.

We seek a functional $F(\{p_i\})$ where $p_i = |a_i|^2$ (the branch weights) satisfying: (1) Permutation invariance — F is symmetric in its arguments; (2) Subdivision additivity — $F(\{p_1, \dots, q \cdot p_k, (1-q) \cdot p_k, \dots, p_n\}) = F(\{p_i\})$ for any $q \in [0,1]$; (3) Normalization — $F(\{p_i\}) = 1$ for all normalized distributions $\sum p_i = 1$.

Proof of the Uniqueness Result

The additivity condition means: if branch k is split into two sub-branches, the total measure is unchanged. Combined with permutation invariance, this constrains F to power-sum functionals $F_r(\{p_i\}) = \sum p_i^r$. Normalization requires $F_r(\{p_i\}) = 1$ for all $\{p_i\}$ with $\sum p_i = 1$. This holds iff $r = 1$, giving $F(\{p_i\}) = \sum p_i = 1$. Since $p_i = |a_i|^2$, we have $F = \sum |a_i|^2 = C_{\text{total}}$.

- The 2-norm is therefore the unique functional of the branch distribution that is permutation-invariant, additive under orthogonal subdivision, and normalized. This converts 'information capacity' from an interpretive label into a theorem.
- Connection to the Holevo Bound
- The Holevo quantity $\chi(E) = H(\{|a_i|^2\})$ bounds accessible classical information from the branch ensemble. Theorem 3 shows the conserved $C_{\text{total}} = 1$ is the normalization constraint that makes $\chi(E)$ well-defined. The three together — Holevo bound, operational

derivation of 2-norm, and Born rule via Busch — mean the framework conserves the measure that generates the Holevo bound.

The additivity condition means: if branch k is split into two sub-branches, the total measure is unchanged. This functional equation, combined with permutation invariance, constrains F to the class of power-sum functionals $F_r(\{p_i\}) = \sum_i p_i^r$. Normalization then requires: for any distribution $\{p_i\}$ with $\sum p_i = 1$, $F_r(\{p_i\}) = 1$. This holds if and only if $r = 1$, giving $F(\{p_i\}) = \sum_i p_i = 1$. Since $p_i = |a_i|^2$, we have $F = \sum_i |a_i|^2 = C_{\text{total}}$.

The 2-norm is therefore the unique functional of the branch distribution that is permutation-invariant, additive under orthogonal subdivision, and normalized to 1 for all quantum states. This converts the 'information capacity' language from an interpretive label into a theorem.

Connection to the Holevo Bound

The Holevo quantity $\chi(E)$ measures how much classical information is accessible from the branch ensemble E . Theorem 3(b) shows that the maximum accessible information is $\log n$ bits for an n -branch superposition, achieved when the Born weights are uniform. The conserved quantity $C_{\text{total}} = 1$ is the normalization constraint that makes $\chi(E)$ well-defined. These three together — the Holevo bound, ICP conservation, and the Born rule via Busch — mean that the ICP conserves the measure that generates the Holevo bound as a functional of the branch distribution.

The Unified Derivation

With all three theorems, the complete argument is:

1. Operational primitives + CDP axioms (Purification, Ideal Compression, Tomographic Locality, Continuous Reversibility) $\rightarrow$ (Theorem 1) $\rightarrow$ Complex Hilbert space H and C^* -algebra $B(H)$.
2. The capacity measure is the unique permutation-invariant, subdivision-additive, normalized functional of the branch distribution $\rightarrow$ (Theorem 3) $\rightarrow C_{\text{total}} = \sum |a_i|^2$.
3. The capacity measure is unitarily invariant and no-signaling-compatible $\rightarrow$ (Theorem 2) $\rightarrow$ The unique consistent conserved scalar is the 2-norm.
4. The 2-norm is a frame function on Hilbert space $\rightarrow$ (Busch's Theorem) $\rightarrow$ The unique probability measure over POVM outcomes is the Born rule: $P(i) = |a_i|^2$.

The Born rule is the unique probability measure consistent with information conservation on complex Hilbert space, independently of interpretive commitments.

4. The Copenhagen–Everett Bridge

The Born rule has kept Copenhagen and Everett apart because each requires the Born rule but cannot ground it from within its own resources. Copenhagen postulates it; Everett derives it from contested rationality axioms. The operational reconstruction resolves this by providing a derivation that is interpretation-neutral.

The derivation proceeds as follows:

1. Operational axioms (Purification, Ideal Compression, Tomographic Locality, Continuous Reversibility) derive complex Hilbert space and the Born rule (Sections 2–3a).

2. Unitary evolution on the derived Hilbert space conserves the 2-norm: $\sum |a_i|^2 = 1$. This conservation is exact, interpretation-independent, and holds across branching, decoherence, and measurement.
3. The 2-norm is non-negative and normalized. By Busch's theorem, the unique probability measure over POVM outcomes is the Born rule: $P(a) = |\langle a|\psi \rangle|^2$.
5. The bridge has three planks:
 6. Copenhagen plank: In each individual branch, wavefunction collapse produces one definite outcome. This satisfies the derived formalism locally — the information in the pre-measurement state is fully transferred to the post-measurement outcome. Copenhagen's operational picture is preserved exactly.
 7. Everett plank: The branch ensemble as a whole preserves the total information. The Born rule weights — now derived from operational axioms rather than postulated — are real frequencies over the ensemble. The branches are physically real, but the measure over them is determined by the derived formalism, not by equal-weight counting.

Derivation plank: The Born rule is the unique measure consistent with the operational axioms on quantum state space. It is not an arbitrary postulate (Copenhagen's problem) nor a consequence of rationality assumptions (Everett's problem). It is a theorem of the formalism both interpretations share.

This resolves the unfalsifiability concern. Copenhagen and Everett make identical predictions not because interpretive questions are meaningless, but because both are interpretations of the same derived formalism. The empirical content lies in the operational axioms themselves — which are testable — not in choosing between interpretations.

Everett plank: The branch ensemble as a whole preserves the total information. The Born rule weights — now derived from ICP rather than postulated — are real frequencies over the ensemble. The branches are physically real, but the measure over them is determined by conservation rather than by equal-weight counting.

ICP plank: The Born rule is the unique measure consistent with information conservation on quantum state space. It is not an arbitrary postulate (Copenhagen's problem) nor a derived consequence of rationality (Everett's problem). It is a physical necessity imposed by conservation.

This resolves the unfalsifiability problem. Copenhagen and Everett have been criticized as empirically indistinguishable. The ICP shows that this indistinguishability is not a weakness but a consequence of a deeper conservation law. They are not competing theories. They are compatible descriptions of the same conserved structure at different levels of analysis.

5. The Uniqueness Argument

Gleason's theorem (1957) establishes that for any separable Hilbert space of dimension ≥ 3 , the unique probability measure satisfying non-negativity and normalization over projective measurements is the Born rule. Busch (2003) extends this to POVMs in all dimensions including dimension 2.

The key mathematical object is the frame function: a map m from unit vectors to non-negative reals such that for every orthonormal basis $\{e_i\}$, $\sum m(e_i) = 1$. Gleason proves every such frame function is $m(v) = \text{Tr}(\rho|v\rangle\langle v|)$ for a unique density operator ρ .

The operational reconstruction contributes a physical derivation, not just an interpretation. The 2-norm is not postulated — it is derived as the unique conserved scalar on the Hilbert space that emerges from the operational axioms. The non-negativity and normalization conditions of Gleason's theorem are satisfied by this derived 2-norm.

An objection: quasi-probability distributions (Wigner, Kirkwood-Dirac) take negative values without violating conservation laws. If negative quasi-probabilities are allowed, why require non-negativity for the Born rule?

The answer: quasi-probability distributions are not probability measures over measurement outcomes. The Wigner function is a phase-space representation; its negativity reflects non-classicality, not outcome probabilities. Neither is a frame function over an orthonormal basis, and neither satisfies the additivity condition Gleason's theorem requires. They are not counterexamples to uniqueness.

The relationship is now precise: let $|\Psi\rangle = \sum a_i|i\rangle$ be the pre-measurement state. The 2-norm $I = \sum |a_i|^2 = 1$ is conserved under unitary evolution. When measurement occurs, the 2-norm is redistributed across branches but its total is unchanged. The Born rule $P(i) = |a_i|^2$ is the local expression of this globally conserved quantity. Non-negativity follows from $|a_i|^2 \geq 0$. Normalization follows from $\sum |a_i|^2 = 1$. Uniqueness follows from Busch's theorem.

6. Probability-Weighted Branch Realism

The operational derivation establishes that the Born rule probabilities are physically necessary — uniquely determined by the axioms, not by interpretive choice. The interpretive question shifts from what the probabilities mean to what physical structure realizes them.

Probability-Weighted Branch Realism (PWBR) answers this. The 2-norm of each branch amplitude is both the conserved informational capacity of that branch and the Born-rule weight of its associated outcome. When the Born rule assigns $P(a) = |a|^2$, this is a physical claim: the informational capacity in branches yielding outcome a is $|a|^2$ of the total conserved measure.

Formally, let B be the set of all branches where measurement M is performed on system S in state $|\psi\rangle$. Let $B(a) \subset B$ be the subset with outcome a . PWBR proposes: $\mu(B(a)) = |\langle a|\psi\rangle|^2$, where μ is the unique Borel probability measure on the branch ensemble derived from the operational axioms — not postulated, but physically necessary.

PWBR occupies the seam between Copenhagen and Everett:

Against strict Everett: Branches are not equally real. A branch with $|a|^2 = 0.001$ is real but carries measure 0.001 — sparse, low-density in branch space. This avoids ontological extravagance while preserving the branch ensemble as physically real.

Against strict Copenhagen: The branch ensemble is physically real, not a calculational fiction. The conserved 2-norm of the universal state is distributed across real branches, and this conserved quantity — not a brute postulate — determines the Born weights.

The bridge: Copenhagen is right about what happens in each branch. Everett is right that branches are physically real. The operational derivation explains why the probability weights have the values they do.

7. Implications for the Discovery Plateau Hypothesis

The operational derivation closes a logical gap in the broader DPH research program. The earlier ICP statement (Varney, 2025a) postulated information conservation but did not specify how the Born rule emerges from non-circular premises. This paper fills that gap: the measure is the Born rule, derived from operational axioms (Purification, Ideal Compression, Tomographic Locality, Continuous Reversibility) via the CDP reconstruction, and the measure-theoretic structure of the branch ensemble is grounded in this derivation. The branch ensemble is real (PWBR); its measure is the Born rule (operational derivation + Busch).

8. Honest Limitations

8. Purification and ideal compression. The CDP reconstruction relies on operational axioms that are independently motivated but not themselves derived from a deeper principle. Purification (every mixed state has a pure-state extension) and ideal compression (redundant degrees of freedom can be removed reversibly) have direct experimental interpretations (no-hiding theorem, Schumacher compression), but closing this gap would require showing they follow from something more fundamental. This remains open work.
9. Quantum structure required. The argument applies to quantum measurements described by POVMs on the Hilbert space derived from the axioms. It does not derive probability measures for hypothetical post-quantum theories that violate the operational primitives (e.g., theories without purification or tomographic locality).
10. Dimension 1. In trivial one-dimensional Hilbert spaces, the normalized state is unique and all outcomes are deterministic. The derivation holds trivially but adds no physical content.
11. The quasi-probability objection. Wigner functions and Kirkwood-Dirac distributions are not frame functions over orthonormal bases and are not counterexamples to uniqueness. Theorem 2 reinforces this: quasi-probabilities violate no-signaling (F3) when treated as physical measures. The deeper question of why Hilbert space itself — and not a more general structure — is the arena of physics is addressed by the CDP axioms but not fully resolved at the level of physical first principles.
12. The bridge unifies rather than chooses. This paper does not argue that Copenhagen is correct or that Everett is correct. It argues that both are interpretations of a formalism in which the Born rule is a theorem of operational axioms. The choice between them remains a matter of which level of description is more useful for a given problem.

9. Conclusion

The Copenhagen–Everett standoff has persisted because the Born rule appeared to require a choice: postulate (Copenhagen) or derive from contested rationality axioms (Everett). The operational reconstruction shows this is a false dilemma. The Born rule — and the Hilbert space

structure it lives on — follows from operational information-theoretic axioms (Purification, Ideal Compression, Tomographic Locality, Continuous Reversibility) that are interpretation-neutral and experimentally testable.

Three theorems close the derivation. Theorem 1 shows Hilbert space and C^* -algebra structure follow from operational primitives plus the CDP axioms, eliminating circularity. Theorem 2 shows the 2-norm is the unique unitarily-invariant, no-signaling-compatible conserved scalar on the derived Hilbert space, making the Born rule forced rather than merely consistent. Theorem 3 grounds the capacity interpretation in a uniqueness result: the 2-norm is the only permutation-invariant, subdivision-additive, normalized functional of the branch distribution.

The bridge resolves the unfalsifiability concern. Copenhagen and Everett make identical predictions because both are interpretations of the same derived formalism. The empirical content lies in the operational axioms — which are testable — not in choosing between interpretations. The framework's physical predictions (e.g., for the DPH program) follow from the derived formalism, not from interpretive commitments.

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